2 Samples: Estimating Difference of Means

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Two populations with means μ_1 and μ_2 Two populations should be indpt
& variances σ_1^2 and σ_2^2 (known variances)
Random samples of sizes n_1 and n_2
We want to assess the difference between μ_1 & μ_2

$\mu_1 - \mu_2$ or $\mu_2 - \mu_1$

$X_1, \ldots X_{n_1}$ | $Y_1, \ldots Y_{n_2}$

E.g. - Miles travelled in car brands A vs B
- Score difference between males vs females
- Avg Q2 marks & avg. Q1 Marks

$\bar{X} - \bar{Y}$

$E(X_i) = \mu_1$; $E(Y_i) = \mu_2$
$V(X_i) = \sigma_1^2$; $V(Y_i) = \sigma_2^2$

$E[\bar{X} - \bar{Y}] = \mu_1 - \mu_2$

$\bar{X} \sim N(\mu_1, \frac{\sigma_1^2}{n_1})$

$V(\bar{X} - \bar{Y}) = \cancel{\sigma_1^2 + \sigma_2^2}$
$= V(\bar{X}) + V(\bar{Y}) - \cancel{2\,cov(\bar{X}, \bar{Y})}^{\,0}$
$= \frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}$

$\bar{Y} \sim N(\mu_2, \frac{\sigma_2^2}{n_2})$

$\sum a_i Y_i \sim N(\ldots)$

$$\frac{\bar{X} - \bar{Y} - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \sim N(0,1)$$

CLT

$$Z = \frac{\bar{X} - \bar{Y} - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \sim N(0,1)$$

$$P(-z_{\alpha/2} < Z < z_{\alpha/2}) = 1 - \alpha$$

Confidence Interval for $\mu_1 - \mu_2$, σ_1^2 and σ_2^2 Known

If $\bar{x}_1$ and $\bar{x}_2$ are means of independent random samples of sizes n_1 and n_2 from populations with known variances σ_1^2 and σ_2^2, respectively, a $100(1-\alpha)\%$ confidence interval for $\mu_1 - \mu_2$ is given by

$$(\bar{x}_1 - \bar{x}_2) - z_{\alpha/2}\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}} < \mu_1 - \mu_2 < (\bar{x}_1 - \bar{x}_2) + z_{\alpha/2}\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}},$$

where $z_{\alpha/2}$ is the z-value leaving an area of $\alpha/2$ to the right.

Exact normality when underlying population is normal

Approximate normality (through CLT) otherwise.

Example 9.10: A study was conducted in which two types of engines, A and B, were compared. Gas mileage, in miles per gallon, was measured. Fifty experiments were conducted using engine type A and 75 experiments were done with engine type B. The gasoline used and other conditions were held constant. The average gas mileage was 36 miles per gallon for engine A and 42 miles per gallon for engine B. Find a 96% confidence interval on $\mu_B - \mu_A$, where μ_A and μ_B are population mean gas mileages for engines A and B, respectively. Assume that the population standard deviations are 6 and 8 for engines A and B, respectively.

$\alpha = 0.04$, $Z_{0.02} = 2.05$

We are 96% confident that the interval (3.43, 8.57) covers the difference in means of gas mileage between engine B and engine A, with engine B having the higher mean.

What about unknown variances?
- if underlying distributions are normal
  or if $n \geq 30$ & non-skewed distribution

$s_1 \approx \sigma_1$ and $s_2 \approx \sigma_2$

- t-distribution

$\underbrace{\bar{X}}\ \ \underbrace{\bar{Y}}\ \ \frac{\bar{X}+\bar{Y}}{2}$

$X_1, \ldots X_{n_1}$ | $Y_1, \ldots Y_{n_2}$

Variances Unknown but Equal

$\sigma_1^2 = \sigma_2^2 = \sigma^2 = \frac{1}{n_a}\sum_{i=1}^{n}(a_i - \bar{a})^2$

$$\boxed{s_p^2 = \frac{\sum(X_i - \bar{X})^2 + \sum(Y_i - \bar{Y})^2}{(n_1 + n_2 - 2)}}$$
$= \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{\ldots}$

Does this mean $s_1^2 = s_2^2 = ??$ X

Variances Unknown but Equal $\sigma_1^2 = \sigma_2^2 = \sigma^2 = \frac{1}{n}\sum_{i=1}^{n}(a_i - \bar{a})^2$ $\boxed{S_p^2 = \frac{\Sigma(x_i-\bar{x})^2 + \Sigma(y_i-\bar{y})^2}{(n_1+n_2-2)} = \frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1+n_2-2}}$

Does this mean $s_1^2 = s_2^2$ = ?? X

$(n_1-1)s_1^2 + (n_2-1)s_2^2$ = ??

$\rightarrow \frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{\sigma^2} \sim \chi^2_{n_1+n_2-2}$

$$\boxed{\frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}} \sim N(0,1)}$$

$\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}$

$$\boxed{\frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{\sigma^2} \sim \chi^2_{n_1+n_2-2}}$$

$$\boxed{\frac{(n-1)S^2}{\sigma^2} \sim \chi^2_{n-1}}$$

Pooled Estimate of Variance

$$S_p^2 = \frac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{n_1+n_2-2}.$$

S_p^2 estimate $\sigma^2 = \sigma_1^2 = \sigma_2^2$

$$\frac{\dfrac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\cancel{\sigma}\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}}{\sqrt{\dfrac{(n_1-1)S_1^2 + (n_2-1)S_2^2}{\cancel{\sigma^2}(n_1+n_2-2)}}}$$

$$\frac{(\bar{X}_1 - \bar{X}_2) - (\mu_1 - \mu_2)}{S_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}}} \sim t_{n_1+n_2-2}$$

Confidence Interval for $\mu_1 - \mu_2$, $\sigma_1^2 = \sigma_2^2$ but Both Unknown

If $\bar{x}_1$ and $\bar{x}_2$ are the means of independent random samples of sizes n_1 and n_2, respectively, from approximately normal populations with unknown but equal variances, a $100(1-\alpha)\%$ confidence interval for $\mu_1 - \mu_2$ is given by

$$(\bar{x}_1 - \bar{x}_2) - t_{\alpha/2}s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}} < \mu_1 - \mu_2 < (\bar{x}_1 - \bar{x}_2) + t_{\alpha/2}s_p\sqrt{\frac{1}{n_1} + \frac{1}{n_2}},$$

where s_p is the pooled estimate of the population standard deviation and $t_{\alpha/2}$ is the t-value with $v = \boxed{n_1 + n_2 - 2}$ degrees of freedom, leaving an area of $\alpha/2$ to the right.

The article "Macroinvertebrate Community Structure as an Indicator of Acid Mine Pollution," published in the *Journal of Environmental Pollution*, reports on an investigation undertaken in Cane Creek, Alabama, to determine the relationship between selected physiochemical parameters and different measures of macroinvertebrate community structure. One facet of the investigation was an evaluation of the effectiveness of a numerical species diversity index to indicate aquatic degradation due to acid mine drainage. Conceptually, a high index of macroinvertebrate species diversity should indicate an unstressed aquatic system, while a low diversity index should indicate a stressed aquatic system.

Two independent sampling stations were chosen for this study, one located downstream from the acid mine discharge point and the other located upstream. For 12 monthly samples collected at the downstream station, the species diversity index had a mean value $\bar{x}_1 = 3.11$ and a standard deviation $s_1 = 0.771$, while 10 monthly samples collected at the upstream station had a mean index value $\bar{x}_2 = 2.04$ and a standard deviation $s_2 = 0.448$. Find a 90% confidence interval for the difference between the population means for the two locations, assuming that the populations are approximately normally distributed with equal variances.

$t_{0.05, 20} = 1.725$

$$\boxed{\mu_1 - \mu_2 > 0}$$

Interpretation $0.593 < \mu_1 - \mu_2 < 1.547$

90% confident that $(.593, 1.547)$ intervals contains the difference of population means for values of species diversity index.

Unknown and Unequal Variances

$$P = \frac{\bar{X}_1 - \bar{X}_2 - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \overset{??}{\sim} t_\nu$$

$$\nu = \left\lfloor \frac{(s_1^2/n_1 + s_2^2/n_2)^2}{\left[\left(\frac{s_1^2}{n_1}\right)^2\left(\frac{1}{n_1-1}\right) + \left(\frac{s_2^2}{n_2}\right)^2\left(\frac{1}{n_2-1}\right)\right]} \right\rfloor$$

Satterthwaite approximation

Confidence Interval for $\mu_1 - \mu_2$, $\sigma_1^2 \neq \sigma_2^2$ and Both Unknown

If $\bar{x}_1$ and s_1^2 and $\bar{x}_2$ and s_2^2 are the means and variances of independent random samples of sizes n_1 and n_2, respectively, from approximately normal populations with unknown and unequal variances, an approximate $100(1-\alpha)\%$ confidence interval for $\mu_1 - \mu_2$ is given by

$$(\bar{x}_1 - \bar{x}_2) - t_{\alpha/2}\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}} < \mu_1 - \mu_2 < (\bar{x}_1 - \bar{x}_2) + t_{\alpha/2}\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}},$$

where $t_{\alpha/2}$ is the t-value with

$$v = \frac{(s_1^2/n_1 + s_2^2/n_2)^2}{[(s_1^2/n_1)^2/(n_1-1)] + [(s_2^2/n_2)^2/(n_2-1)]}$$

degrees of freedom, leaving an area of $\alpha/2$ to the right.

$$\frac{(\bar{x}_1 - \bar{x}_2) - (\mu_1 - \mu_2)}{\sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}} \sim t_\nu$$

ν is an "estimate" of degrees of freedom

In summary, all CIs for mean look like

point estimate $\pm$ $t_{\alpha/2}\,\widehat{\text{s.e.}}$(point estimate)

or

point estimate $\pm$ $z_{\alpha/2}$ s.e.(point estimate).

2 Samples: Paired observations

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When two populations are not independent
& the variances of two populations are not necessarily equal

Marks in Quiz 1 | Marks in Quiz 2

$\boxed{\mu_1 - \mu_2 \text{ or } \mu_2 - \mu_1}$

Paired observation

(X_i, Y_i) dependence $\quad d_i = Y_i - X_i$

(X_1, Y_1) — S1

(X_2, Y_2) — S2 indpt.

$\bar{X} - \bar{Y}$

- Weight before and after Diet 1

differences $d_1, d_2, \ldots d_n$

$D_i = X_{1i} - X_{2i}$

Confidence Interval for $\mu_D = \mu_1 - \mu_2$ for Paired Observations

If $\bar{d}$ and s_d are the mean and standard deviation, respectively, of the normally distributed differences of n random pairs of measurements, a $100(1-\alpha)\%$ confidence interval for $\mu_D = \mu_1 - \mu_2$ is

$$\bar{d} - t_{\alpha/2}\frac{s_d}{\sqrt{n}} < \mu_D < \bar{d} + t_{\alpha/2}\frac{s_d}{\sqrt{n}},$$

where $t_{\alpha/2}$ is the t-value with $v = n - 1$ degrees of freedom, leaving an area of $\alpha/2$ to the right.

Example 9.13: A study published in *Chemosphere* reported the levels of the dioxin TCDD of 20 Massachusetts Vietnam veterans who were possibly exposed to Agent Orange. The TCDD levels in plasma and in fat tissue are listed in Table 9.1.

Find a 95% confidence interval for $\mu_1 - \mu_2$, where μ_1 and μ_2 represent the true mean TCDD levels in plasma and in fat tissue, respectively. Assume the distribution of the differences to be approximately normal.

Table 9.1: Data for Example 9.13

Veteran	TCDD Levels in Plasma	TCDD Levels in Fat Tissue	d_i	Veteran	TCDD Levels in Plasma	TCDD Levels in Fat Tissue	d_i
1	2.5	4.9	−2.4	11	6.9	7.0	−0.1
2	3.1	5.9	−2.8	12	3.3	2.9	0.4
3	2.1	4.4	−2.3	13	4.6	4.6	0.0
4	3.5	6.9	−3.4	14	1.6	1.4	0.2
5	3.1	7.0	−3.9	15	7.2	7.7	−0.5
6	1.8	4.2	−2.4	16	1.8	1.1	0.7
7	6.0	10.0	−4.0	17	20.0	11.0	9.0
8	3.0	5.5	−2.5	18	2.0	2.5	−0.5
9	36.0	41.0	−5.0	19	2.5	2.3	0.2
10	4.7	4.4	0.3	20	4.1	2.5	1.6

Source: Schecter, A. et al. "Partitioning of 2,3,7,8-chlorinated dibenzo-p-dioxins and dibenzofurans between adipose tissue and plasma lipid of 20 Massachusetts Vietnam veterans," *Chemosphere*, Vol. 20, Nos. 7–9, 1990, pp. 954–955 (Tables I and II).

$t_{0.025, 19} = 2.9773$

$\bar{d} = -0.87$

$$s_d = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(d_i - \bar{d})^2} = \sqrt{\frac{168.4220}{19}} = 2.9773.$$

Ex.1 ① # of rooms (X), area of flat (Y)
- we should be able to compare means

② overall marks of Roki ($\bar{X}$), Yash ($\bar{Y}$)

$X_1, \ldots X_n$

Sub1, ... Sub5   Sub1, ... Sub5

$(X_1, Y_1) \Rightarrow$ (Sub1 (Rohi), Sub1 (Yash)) — indpt.
(Sub2 " , Sub2 ")

⋮

(3) avg. income of h, w
(h_1, w_1) — C_1 — indpt
(h_2, w_2) — C_2

(4) sales of mangoes before & after a promotion event at supermarket
— dependent

(1b) area of houses in different neighborhood.
indpt.